

Question 1

Fig. 1.1 below shows nuclear division occurring in an animal cell.

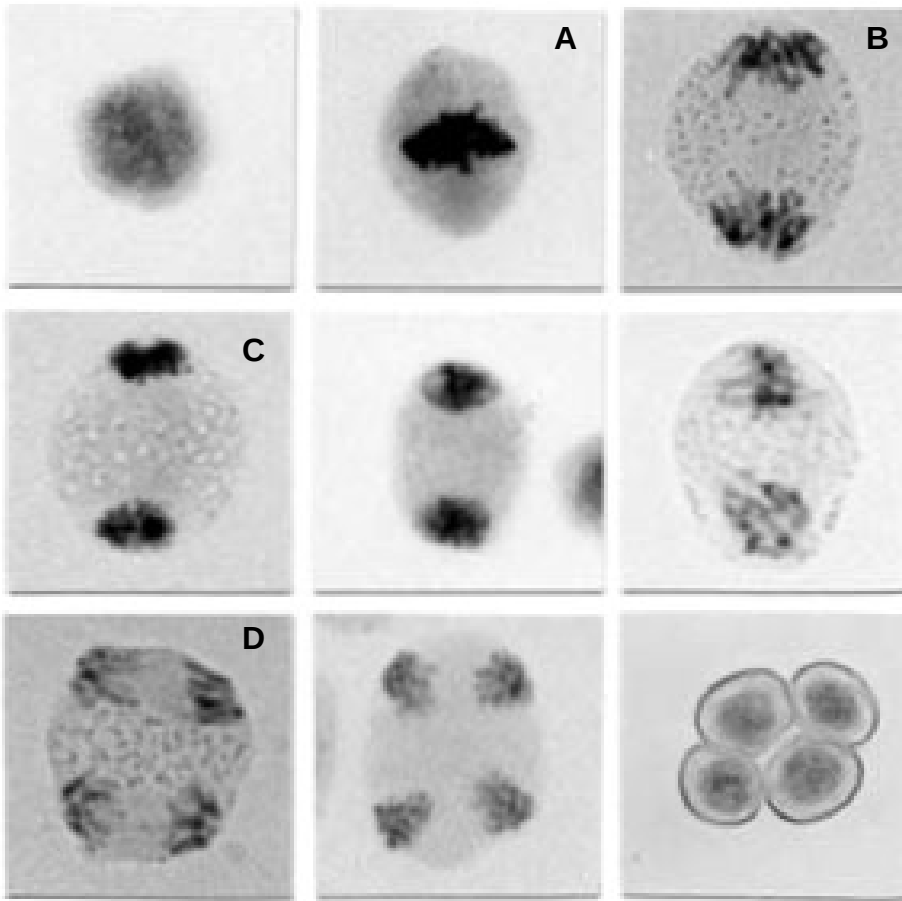


Fig. 1.1

a)

i) Identify stages A to D.

[2]

A

B

C

D

ii) State the type of nuclear division shown in Fig. 1.1

[1]

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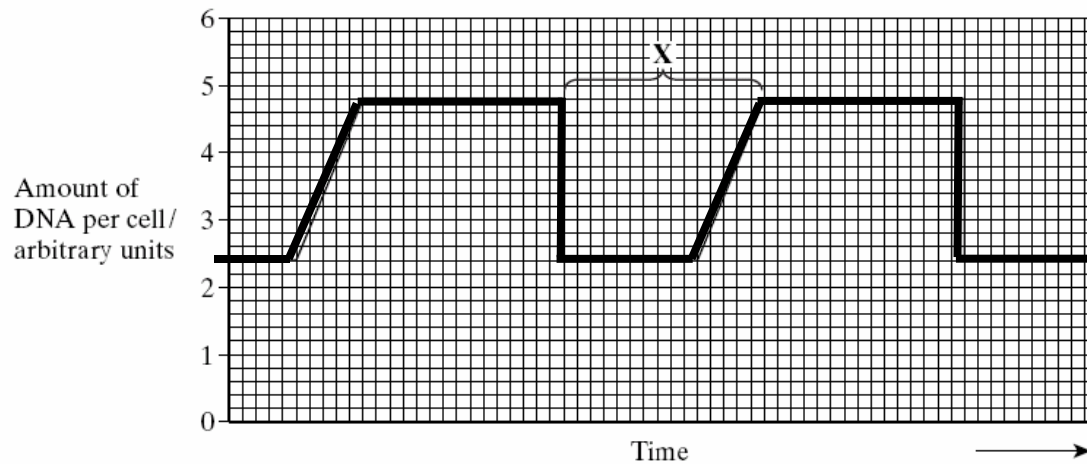
iii) Explain your answer to part (ii).

[1]

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b) The graph below shows the relative amounts of DNA per cell during two successive cell divisions in a plant.



i)) Describe the events occurring at X.

[3]

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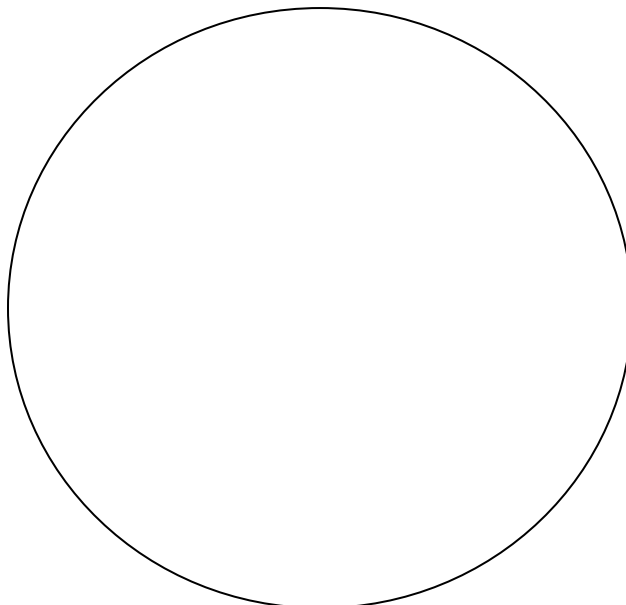
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c) Draw the behaviour of the chromosomes undergoing anaphase II during nuclear division in the cell given below where ($2n = 6$)

[3]



[total: 10]

Question 2

a) Protein synthesis is necessary for the survival of cells where important proteins are synthesised. Protein synthesis involves both the process of transcription and translation. Any disruption to these processes may cause the cell to be dysfunctional.

i) Describe the importance of the DNA-binding domain and protein-binding domain of transcription factors. [2]

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ii) A retrotransposon (a 'jumping gene' containing non-coding DNA) was accidentally inserted at the TATA region in the promoter of a gene. Explain how this would affect transcription of that gene. [2]

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iii) State precisely where in the cell translation occurs. [1]

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iv) Describe how the structure of the ribosome is adapted for translation. [3]

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v) A mutation was found in a gene that codes for the capping enzyme complex. Explain how this mutation would affect translation. [2]

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[total: 10]

Question 3

a) Describe 2 differences between the structure of prokaryotic and eukaryotic genomes. [2]

Prokaryotic genome	Eukaryotic genome

Figure 3.1 below represents the processing of pre-messenger RNA.

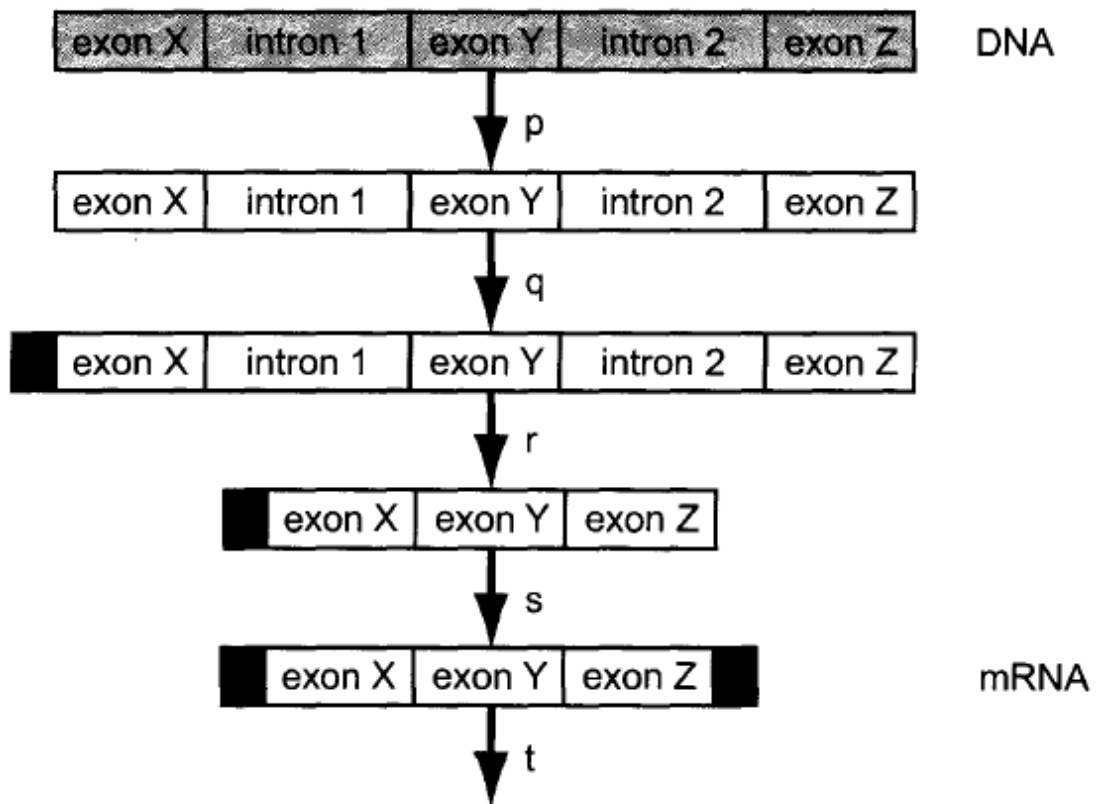


Fig. 3.1

b) Describe the events r to t.

[5]

r:

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s:

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t:

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c) Fig 3.2 shows the action of telomerase enzyme.

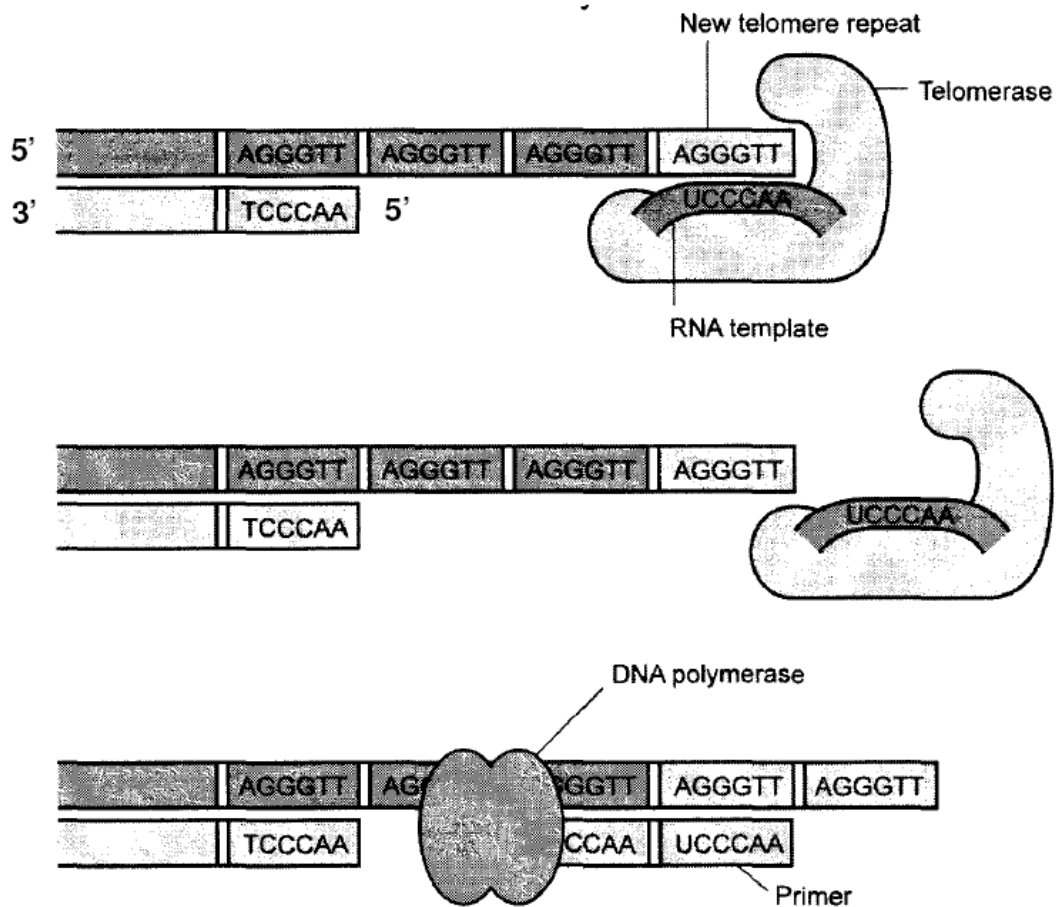


Fig. 3.2

i) State whether the new telomere repeat is added to the daughter or parent DNA strand. Explain your answer. [2]

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ii) Explain why telomerase activity is essential in some unicellular organisms. [1]

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[total: 10 marks]

Question 4

a) Pure-breeding fruit flies, *Drosophila melanogaster*, with wild-type bodies and vestigial wings were crossed with pure-breeding flies with black bodies and wild-type wings. The F1 all had wild-type bodies and wild-type wings.

A test cross was then carried out with F1.

i) State the ratio of phenotypes expected in a dihybrid cross such as this test cross. [1]

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The actual results from this cross were as follows:

Wild-type body, vestigial wings	1252
Wild-type body, wild type wings	254
Black body, vestigial wings	248
Black body, wild-type wings	1246

ii) Draw a genetic diagram to show how this results. [4]

Use symbols: R – wild-type body r – black body
 T – wild-type wings t – vestigial wings

iii) Explain the discrepancy between the expected and actual results.

[4]

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b) In another experiment, another phenotype, eye colour, was examined together with wing shape. Pure-breeding *Drosophila melanogaster* with red eyes and vestigial wings were crossed with pure-breeding flies with black eyes and wild-type wings. The F1 all had black eyes and wild type wings. Females of F1 were then crossed with males of F1.

The experimental results of the F2 generation are as below:

F2 generation

Black eyes and wild-type wings	510
Black eyes and vestigial wings	173
Red eyes and wild-type wings	165
Red eyes and vestigial wings	52

i) Use the chi-square (χ^2) test and the table of probability shown in table 4.1 to find the probability of the results of the F2 generation departing significantly from expectation. Show your working.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'
 ν = degrees of freedom
 c = number of classes

O = observed 'value'
E = expected 'value'

Distribution of χ^2

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$\chi^2 =$ Probability =[4]

[total: 13]

Question 5

The pathway of carbon during the reactions of photosynthesis was discovered using the unicellular green algae *Chlorella*. The *Chlorella* culture was supplied with $^{14}\text{CO}_2$ and illuminated with varying length of time. Some algae were removed at intervals and immediately killed by boiling in alcohol. This provided an extract which was examined using chromatographic techniques and any radioactive compounds present showed up on X-ray film as blackened areas.

After 15 seconds of illumination, the extract produced three prominent blackened areas which were due to the presence of three compounds:

- i. glycerate phosphate (GP/ PGA)
- ii. ribulose biphosphate (RuBP)
- iii. glyceraldehyde-3-phosphate (G3P)

a) In a second series of experiments the algae were illuminated in the presence of $^{14}\text{CO}_2$ for 15 seconds. Illumination of the algae continues for a further 1 minute after all $^{14}\text{CO}_2$ present had been removed. Some algae were then removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

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b) In another series of experiments, the algae were illuminated in the presence of $^{14}\text{CO}_2$ for 15 seconds and then placed in the dark for 1 min in the presence of $^{14}\text{CO}_2$. Some algae were removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

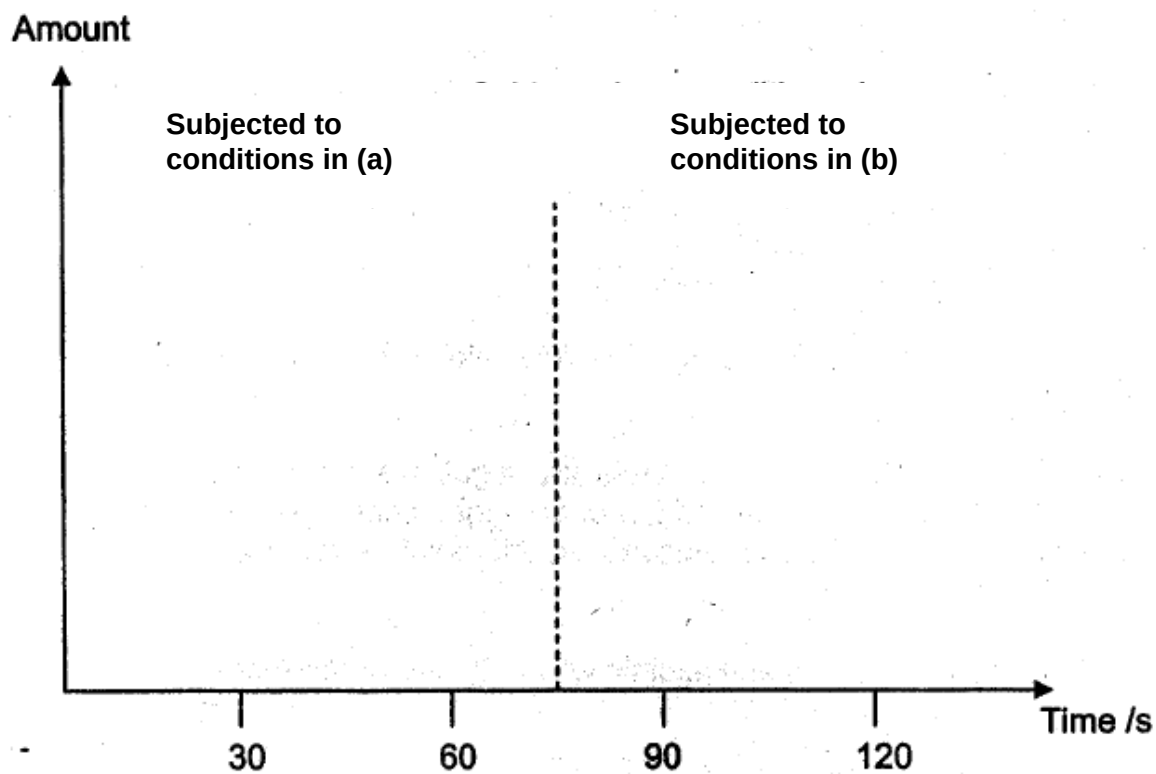
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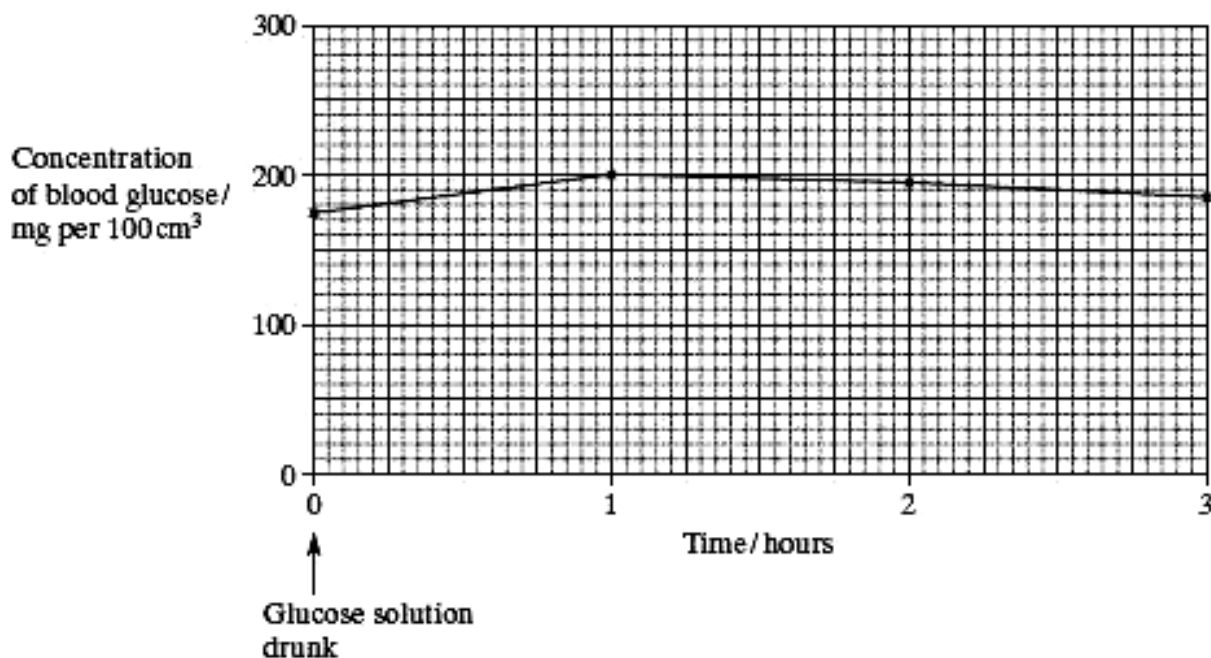
c) On the graph below, show how the amounts of the labelled compounds named in part a) and b) will change if the algae is *first* subjected to the conditions in (a) *then* to conditions in (b). [3]



[total: 7]

Question 6

Type 1 diabetes is often due to the inability to produce insulin. In an investigation, a man with type 1 diabetes drank a glucose solution. The concentration of glucose in his blood was measured at regular intervals. The results are shown in the graph below.



- a) Suggest **two** reasons why the concentration of glucose decreased after 1 hour even though this man's blood contained no insulin. [2]

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- b) The investigation was repeated on a man who did not have diabetes. The concentration of glucose in his blood before drinking the glucose solution was 80 mg per 100 cm³. Sketch a curve on the graph above to show the results you would expect. [1]

- c) Explain how the blood insulin is homeostatically regulated in a normal individual. [3]

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Figure 6.2 shows the normal sequence of cell signalling events when insulin reaches its target cell and binds to an insulin receptor resulting in an increase in the number of glucose channels in the cell surface membrane.

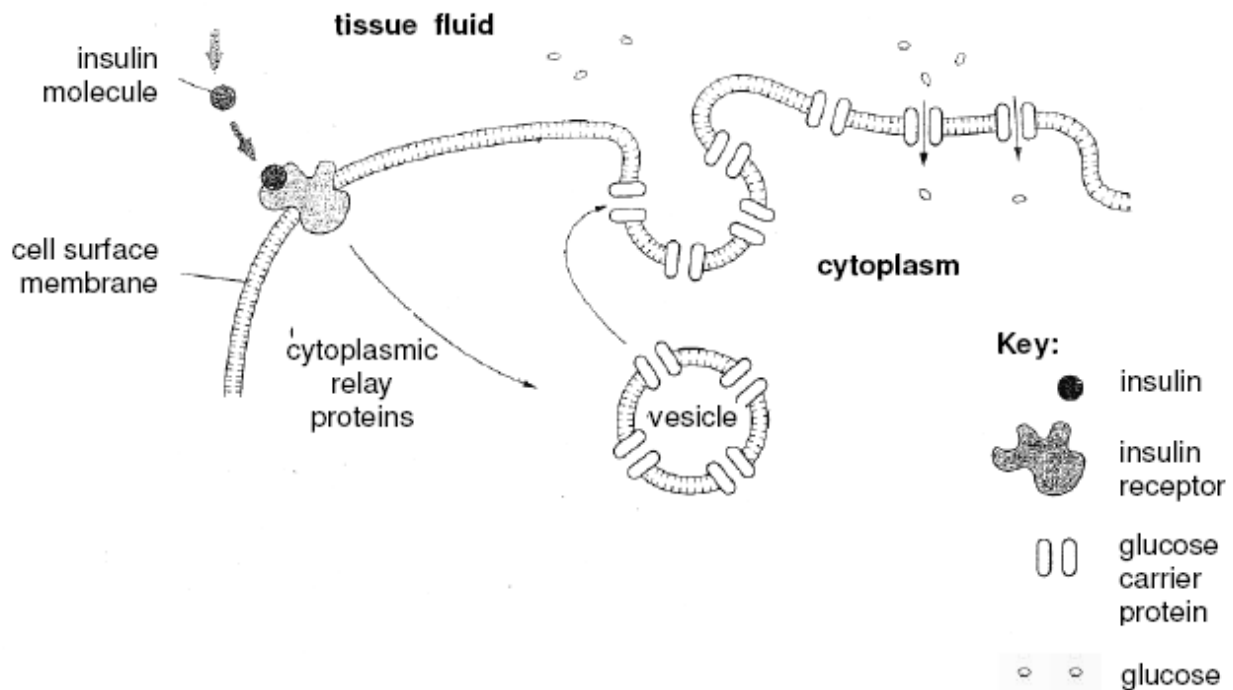


Fig. 6.2

d) With reference to Fig 6.2 and using your biological knowledge, describe the events that ultimately result in an increase in glucose uptake. [4]

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[total: 10]

Question 7

Figure 7.1 below shows part of two nerve fibres and a synapse. The figures indicate the value in mV of the potential across the membrane between the cytoplasm of the fibres and the extracellular fluid at intervals along each fibre.

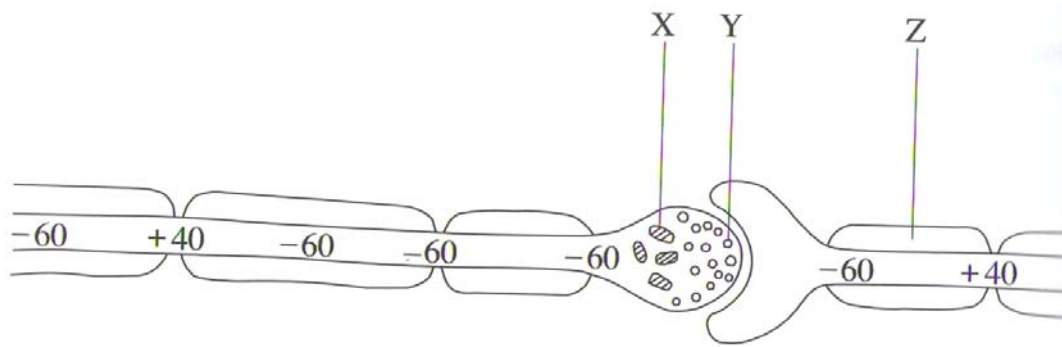


Fig. 7.1

ai) Draw a circle around one region of the diagram where an action potential exists. Explain your choice. [2]

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ii) By means of an arrow on the diagram, indicate the direction in which action potentials would normally travel along these fibres. Explain why the transmission of impulse is uni-directional across two nerve cells. [2]

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iii) Identify structures X and Y, and explain how each is involved in the transmission of nerve impulse. [1]

Structure X:

Explanation:

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Structure Y: [2]

Explanation:

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b) Figure 7.2 shows a normal nerve-muscle junction.

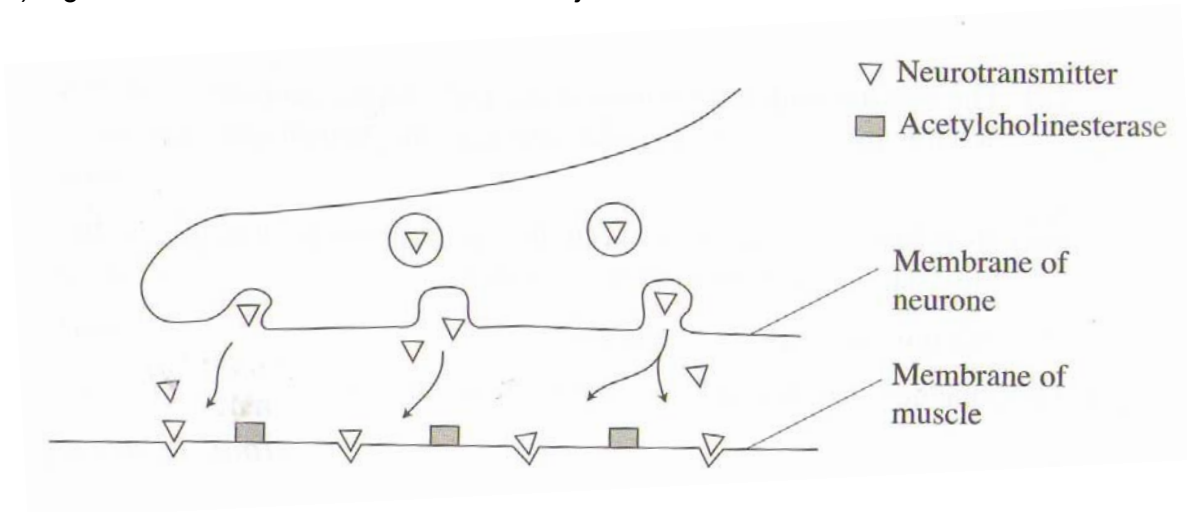


Fig. 7.2a

Using the evidence from Fig. 7.2a and Fig. 7.2b, briefly describe and explain the effect on muscle activity of i) organophosphates and ii) botulin toxin. [3]

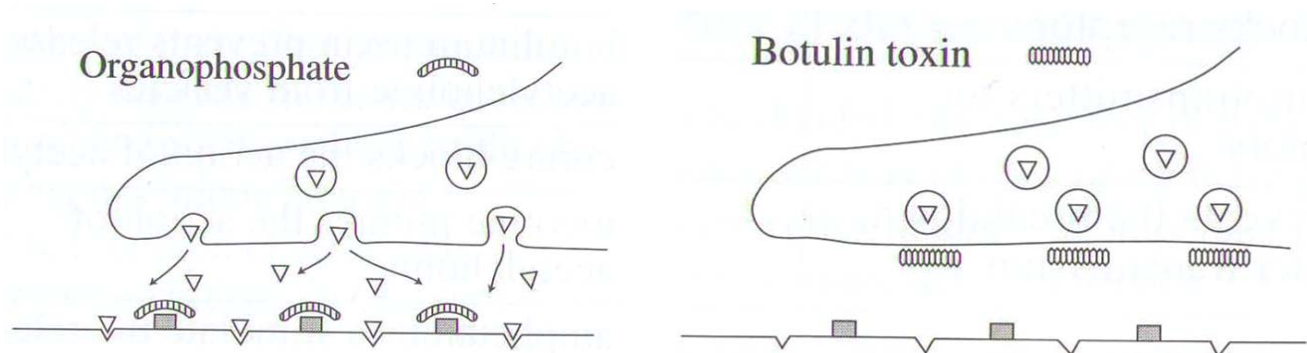


Fig. 7.2b

i) organophosphate:

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ii) botulin toxin

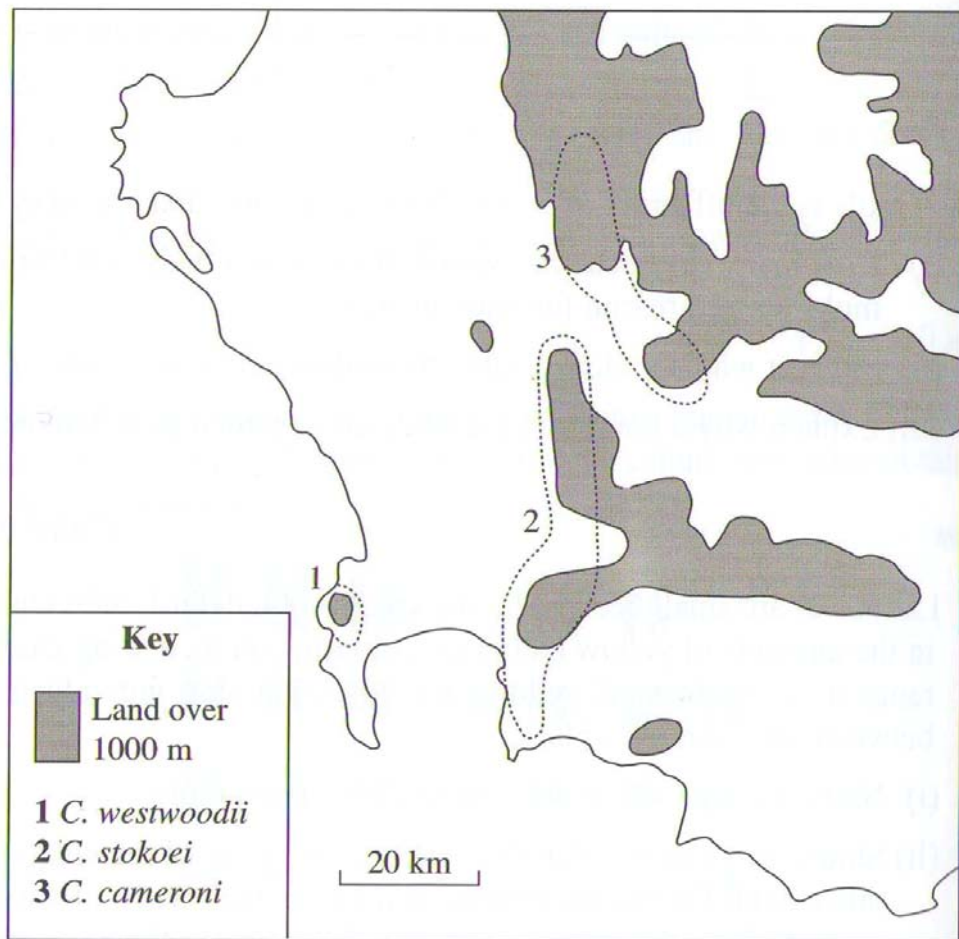
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[total: 10]

Question 8

The beetles belonging to the genus *Colophon* are unable to fly and are found on hilltops in South Africa. The dotted lines on the map show the distribution of three species.



a) Suggest an evolutionary explanation for each of the following statements.

i) All of these beetles are **similar** general appearance.

[1]

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ii) There are **differences** between the species of *Colophon* found in the three areas.

[3]

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[illegible]

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[total: 10]
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Section B
Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

9

- a) With reference to a named virus and its reproductive cycle, outline how specialized transduction could result in a stable transfer of genes into a recipient bacterium. [12]
- b) Explain, with examples, how viral infections could cause diseases in animals. [5]
- c) Hepatitis A is a disease caused by a virus which can permanently damage the liver and pancreas. Suggest why the virus causes damage only to some types of human cells. [3]

OR

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- a) Explain how natural selection may bring about evolution. [7]
- b) Describe the mechanisms that can cause changes in allele frequencies in a population. [6]
- c) Explain, with appropriate examples, how recessive alleles may be preserved in a natural population. [7]